

Name: _____



DANE BANK

An Anglican School for Girls

YEAR 12

Mathematics Extension 2

Assessment Task # 2

Assessment Week April 2007

Time Allowed: 2 hours Plus 5 Minutes Reading Time

Outcomes tested: E2, HE2, E3, E4, E9, HE4, HE6, HE7, E7

Weighting: 20 % Maximum Marks 80

DIRECTIONS TO CANDIDATE:

- Attempt all questions.
- Use the BOOKLETS provided.
- Write using blue or black pen.
- Start each question in a New Booklet.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- All questions are of equal value. Not all parts are of equal value.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or badly arranged work.
- The format of the paper does not necessarily reflect that of the HSC.

Question 1. Start a New Booklet.

16 Marks

Marks

- a) Find $\int x^2 e^{1-x^3} dx$

1

- b) Using the substitution $u=6-x$, evaluate

$$\int_1^6 x \sqrt{6-x} dx$$

3

- c) (i) Differentiate $y=7^x$ with respect to x .

2

- (ii) Hence, or otherwise, find $\int 7^x dx$

1

- d) Find the gradient of the curve $x^3 + 4xy + y^2 = 13$ at the point(s) where $x = 1$.

4

- e) (i) $x=3\sec\theta$ and $y=5\tan\theta$ are the parametric equations of a hyperbola.

Find the cartesian equation of the hyperbola.

1

- (ii) Write the equations of the asymptotes.

1

- (iii) Draw a diagram, with clear labelling, to show the relevance of the numbers 3 and 5.

2

- (iv) State the equation of the auxiliary circle.

1

Question 2. Start a New Booklet.

16 Marks

Marks

- a) Let
- $z = 3 + 2i$
- and
- $w = 1 - i$

Find, in the form $x + iy$,

(i) $z w$

1

(ii) $\frac{z}{w}$

2

- b) (i) Express
- $1 - i$
- in modulus-argument form.

2

(ii) Express $(1 - i)^5$ in modulus-argument form.

2

(iii) Hence, express $(1 - i)^5$ in the form $x + iy$.

1

c) The ellipse E has the equation $\frac{x^2}{100} + \frac{y^2}{75} = 1$.

- (i) Sketch the curve
- E
- , showing on your diagram the coordinates of the foci and equation of each directrix.

3

(ii) Find the equation of the normal to the ellipse at the point $P(5, 7.5)$

3

(iii) Find the equation of the circle that is tangential to the ellipse at P and $Q(5, -7.5)$

2

Question 3. Start a New Booklet.

16 Marks

Marks

- a) The polynomial $A(x) = x^4 + ax^2 + bx + 36$ has a double root at $x = 2$.

Find the values of a and b .

3

- b) A is the point on the Argand Diagram which represents the complex number $z = 3 + i$ and B is the point which represents $-z$.

- (i) Draw a diagram showing the positions of A and B .

1

- (ii) P and Q are points such that figure $APBQ$ is a rhombus whose area is 150 units².

Find the complex numbers represented by P and Q .

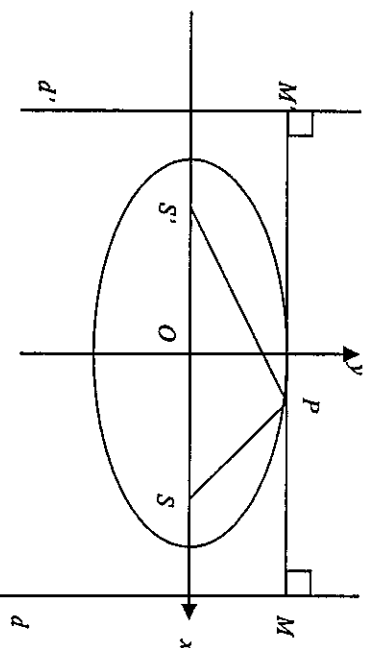
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(Question 3 Continued on next page)

(Question 3 continued)

Marks

c) (i)



Use the diagram above, and the definition of an ellipse, to prove that

$$PS + PS' = 2a$$

2

- (ii) Draw a number line to illustrate the meaning of $|x - 3| = 4$ and complete the statement: “The distance of x from”

1

Use Parts (i) and (ii) to answer the following.

The equation $|z + 3 + i| + |z + 3 + 9i| = 10$ represents an ellipse in the Argand Plane.

- (iii) Sketch the ellipse on the Argand Plane showing: the foci, the end-points of the major and minor axes, the centre of the ellipse.

4

- (iv) Write down the range of values of $\arg z$ for complex numbers z corresponding to points on the ellipse.

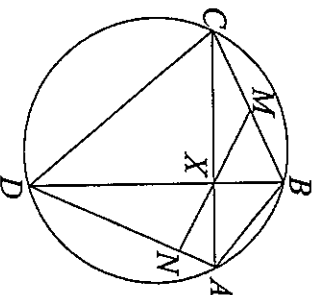
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Question 4. Start a New Booklet.

16 Marks

Marks

a)



$ABCD$ is a cyclic quadrilateral. The diagonals AC and BD intersect at right angles at X . M is the midpoint of BC . MX produced meets AD at N .

Copy the diagram above into your own booklet showing the above information.

1

2

- (i) Explain why BC is the diameter of the circle which can be drawn through points B , C and X .

5

- (ii) Show that $\angle MBX = \angle MXB$.

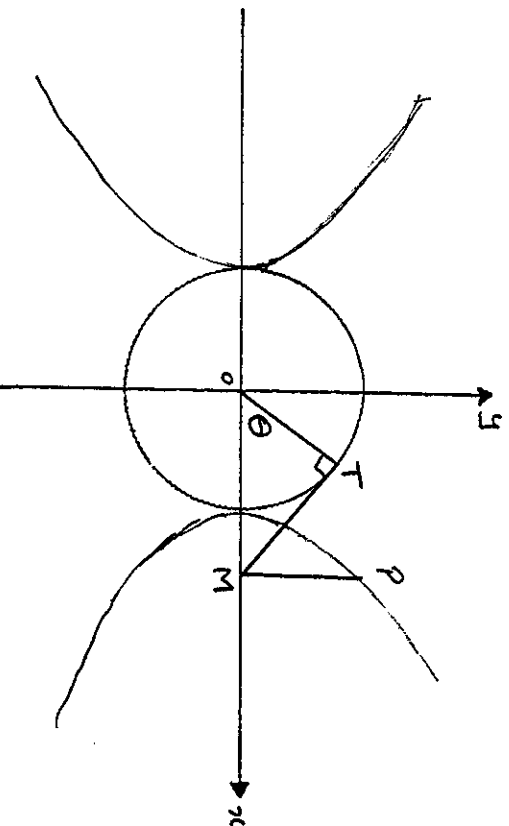
- (iii) Show that MN is perpendicular to AD .

(Question 4 Continued on next page)

(Question 4 continued)

Marks

b)



The figure shows the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and the circle $x^2 + y^2 = a^2$, where $a, b > 0$.

The point T lies on the circle with $\angle TOM = \theta$, where $0 < \theta < \frac{\pi}{2}$. The tangent to the circle at T meets the x -axis at M , MP is perpendicular to the x -axis and P is a point on the hyperbola in the first quadrant.

(i) Show that P has coordinates $(a \sec \theta, b \tan \theta)$.

2

(ii) Suppose that Q is a point on the hyperbola with coordinates $(a \sec \theta, b \tan \theta)$.
If $\theta + \phi = \frac{\pi}{2}$, and $\theta \neq \frac{\pi}{4}$, show that the chord PQ has equation

$$ay = b (\cos \theta + \sin \theta) x - ab$$

3

(iii) Show that every such chord PQ passes through a **fixed** point and find the coordinates of this point.

1

(iv) Show that as θ approaches $\frac{\pi}{2}$ the chord PQ approaches a line parallel to an asymptote.

2

Question 5. Start a New Booklet.

16 Marks

Marks

- a) Suppose that
- b
- and
- d
- are real numbers and
- $d \neq 0$
- . Consider the polynomial

$$P(z) = z^4 + bz^2 + d$$

The polynomial has a double root α .

- (i) Prove that
- $P'(z)$
- is an odd function.

2

- (ii) Prove that
- $-\alpha$
- is also a double root of
- $P(z)$
- .

1

- (iii) Prove that
- $d = \frac{b^2}{4}$

2

- (iv) For what values of
- b
- does
- $P(z)$
- have a double root equal to
- $\sqrt{3}i$
- ?

1

- (v) For what values of
- b
- does
- $P(z)$
- have real roots?

2

- b) (i) Show that for all values of
- A
- and
- B
- ,

$$\sin(A+B) - \sin(A-B) = 2 \cos A \sin B$$

2

- (ii) Use the method of Mathematical Induction to show that for all positive integers,
- n
- ,

$$\cos x + \cos 2x + \cos 3x + \dots + \cos nx = \frac{\sin(n + \frac{1}{2})x - \sin \frac{1}{2}x}{2 \sin \frac{1}{2}x}$$

6

----- End of Test -----

Question	Your Mk	Max Mk	Topic	Outcomes
1 a)		1	Harder ext 1-integration	HE6
1 b)		3	Harder ext 1-integration	HE6
1 c) (i)		2	Harder ext 1-diff'n integration	HE6
1 c) (ii)		1	Harder ext 1-diff'n integration	HE6
1 d)		4	Conics-implicit diff'n	E4
1 e) (i)		1	Conics-elim of parameters	E4
1 e) (ii)		1	Conics-knowledge of terms	E4
1 e) (iii)		2	Conics-understanding of a and b	E4
1 e) (iv)		1	Conics-knowledge of terms	E4
		16		
2 a) (i)		1	Complex Nos-manipulating algebra	E3
2 a) (ii)		2	Complex Nos-manipulating algebra	E3
2 b) (i)		2	Complex Nos-knowing terms	E3
2 b) (ii)		2	Complex Nos-De Moivre's rule	E3
2 b) (iii)		1	Complex Nos-mod/arg→cartesian	E3
2 c) (i)		3	Conics-knowledge of terms	E4
2 c) (ii)		3	Conics-diff'n equation normal	E4
2 c) (iii)		2	Conics-problem, language	E4
		16		
3 a)		3	Polys-finding a and b	PE3 E4
3 b) (i)		1	Complex nos-plotting pts	E3
3 b) (ii)		4	Complex nos-geometry of	E3 HE7 E9
3 c) (i)		2	Conics-proof	E3
3 c) (ii)		1	Harder Ext 1- Geometry of distance	H9
3 c) (iii)		4	Conics-concept of defn & posn foci	E3 E9
3 c) (iv)		1	Complex nos-argument	E3 E9
		16		
4 a) (i)		1	Harder Ext 1-circle geometry	PE3
4 a) (ii)		2	Harder Ext 1-angles & isostrangles	PE3
4 a) (iii)		5	Harder Ext 1-angle sum triangle	PE3
4 b) (i)		2	Conics-concept aux.circle	E4
4 b) (ii)		3	Conics-equation of chord	E4
4 b) (iii)		1	Conics-concept of "fixed pt"	E3 E4
4 b) (iv)		2	Conics-asymptotes and limits	E3 E4
		16		
5 a) (i)		2	Polynomials-odd and even fns	E4
5 a) (ii)		1	Polys-concept of roots in odd fns	E4 HE7 E9
5 a) (iii)		2	Polynomials-use of data	E4 HE7 E9
5 a) (iv)		1	Polys-finding b given root	E4 HE7 E9
5 a) (v)		2	Polys-concept of real roots	E4 HE7 E9
5 b) (i)		2	Harder Ext 1-trig relnsps	HE2 E2
5 b) (ii)		6	Harder Ext 1-Mathematical Indn	E2 HE2
		16		
Total		80		

**Year 12 Outcomes
Mathematics Extension 2
4th April, 2007 Assessment # 2**

Name: _____

[illegible]

Polynomials						
3 a)	5a) (i)	5a) (ii)	5a) (iii)	5a) (iv)	5a) (v)	Tot
3	2	1	2	1	2	11

[illegible][illegible]

2007 Mathematics Extension 2 Assessment #2

Q1 a) $\int x^2 e^{1-x^3} dx = -\frac{1}{3} \int -3x^2 e^{1-x^3} dx$

$= -\frac{1}{3} e^{1-x^3} + K$ (1 mark)

b) $\int_1^6 x \sqrt{6-x} dx = \int_5^0 (6-u) u^{\frac{1}{2}} (-du)^{\frac{1}{2}}$ Put $u = 6-x \Rightarrow x = 6-u$

$du = -1 dx$

$= -\int_5^0 6u^{\frac{1}{2}} - u^{\frac{3}{2}} du$

when $x=1$, $u=5$
 $x=6$, $u=0$

$= -\left[\frac{6u^{\frac{3}{2}}}{\frac{3}{2}} - \frac{u^{\frac{5}{2}}}{\frac{5}{2}} \right]_5^0$

$= -\left[(0) - \left(\frac{2 \times 6 \times 5^{\frac{3}{2}}}{3} - \frac{2 \times 5^{\frac{5}{2}}}{5} \right) \right]$

$= -\left[-\left(4 \times 5\sqrt{5}^3 - \frac{2 \times 5\sqrt{5}^5}{5} \right) \right]$

$= \left[20\sqrt{5} - \frac{2 \times 25\sqrt{5}}{5} \right]$

$= 20\sqrt{5} - 10\sqrt{5}$

$= 10\sqrt{5}$

c) (i) $y = 7^x$

$\therefore \log_7 y = x$

ie $x = \frac{\log_e y}{\log_e 7}$

$\frac{dx}{dy} = \frac{1}{\log_e 7} \cdot \frac{1}{y} \Rightarrow \frac{dy}{dx} = y \log_e 7$
 $= 7^x \log_e 7$

(ii) Hence, $\int 7^x dx = \frac{1}{\log_e 7} \int \ln 7 \cdot 7^x dx$

$= \frac{1}{\ln 7} 7^x + K$

Ans: $\frac{1}{\ln 7} 7^x + K$

d)

$$x^3 + 4xy + y^2 = 13$$

$$\text{when } x=1, \quad 1 + 4y + y^2 = 13$$

$$y^2 + 4y - 12 = 0$$

$$(y+6)(y-2) = 0$$

$$\text{we } y = -6 \text{ or } y = 2$$

$$\text{Points are } (1, -6) \text{ or } (1, 2)^{\frac{1}{2}}$$

$$\text{Now, } 3x^2 + y \cdot 4 + 4x \cdot \frac{dy}{dx} + 2y \frac{dy}{dx} = 0. \quad \frac{1}{2}$$

$$\frac{dy}{dx}(4x + 2y) = -3x^2 - 4y$$

$$\frac{dy}{dx} = \frac{-3x^2 - 4y}{4x + 2y} \quad \frac{1}{2}$$

$$\text{when } x=1, y=-6$$

$$\frac{dy}{dx} = \frac{-3 \times 1 - 4 \times -6}{4 \times 1 + 2 \times -6}$$

$$= \frac{-3 + 24}{4 - 12}$$

$$= \frac{21}{-8}$$

$$= -\frac{21}{8}$$

$$\text{when } x=1, y=2$$

$$\frac{dy}{dx} = \frac{-3 \times 1 - 4 \times 2}{4 \times 1 + 2 \times 2}$$

$$= \frac{-3 - 8}{4 + 4}$$

$$= -\frac{11}{8}$$

e) (i)

$$x = 3 \sec \theta, \quad y = 5 \tan \theta$$

$$\frac{x^2}{9} = \sec^2 \theta, \quad \frac{y^2}{25} = \tan^2 \theta$$

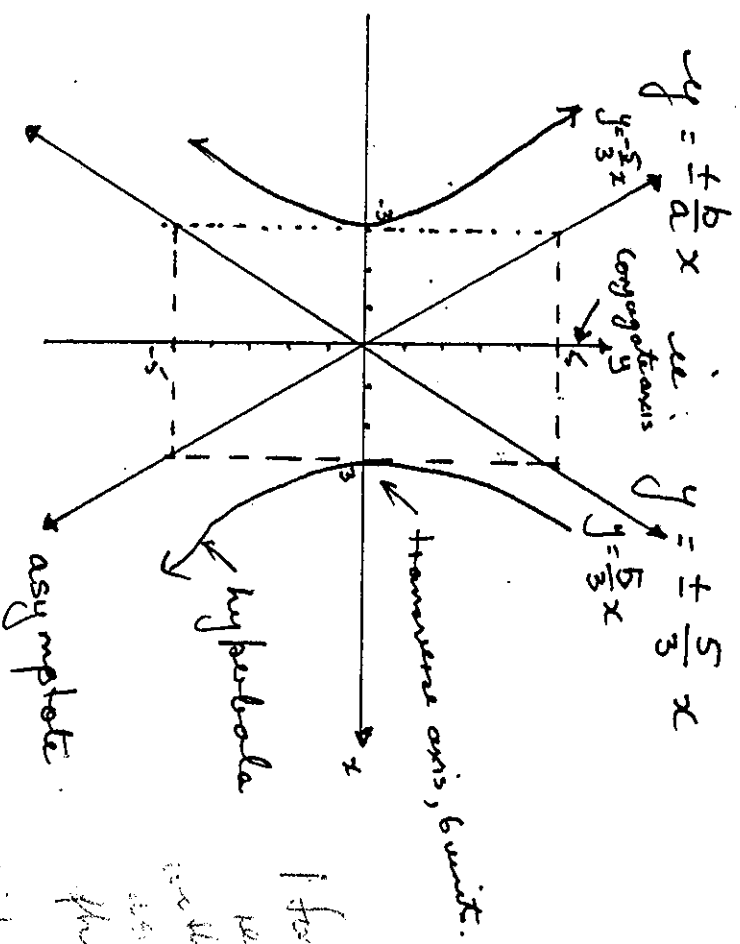
$$\text{Now } \sin^2 \theta + \cos^2 \theta = 1$$

$$\therefore \tan^2 \theta + 1 = \sec^2 \theta$$

$$\text{we } \frac{y^2}{25} + 1 = \frac{x^2}{9}$$

$$\therefore \frac{x^2}{9} - \frac{y^2}{25} = 1 \quad \text{is Cartesian form}$$

(ii) Equations of asymptotes are



(iii)

(iv) Auxiliary circle is $x^2 + y^2 = 9$

1 for vertex
2 for asymptotes
3 for hyperbola
4 for conjugate axis
5 for transverse axis

Question 2.

a) (i) $z^3 = (3+2i)(1-i)$

$$= 3 - 3i + 2i - 2i^2$$

$$= 3 + 2 - i$$

$$= 5 - i$$

(ii)

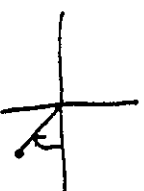
$$\frac{3}{w}$$

$$= \frac{3+2i}{1+i\frac{1}{\sqrt{2}}} \times \frac{1-i\frac{1}{\sqrt{2}}}{1-i\frac{1}{\sqrt{2}}}$$

$$= \frac{3-3i+2i-2i^2}{1-i^2}$$

$$= \frac{5-i}{2}$$

$$= \frac{5}{2} - \frac{1}{2}i$$



b) (i) $|1-i| = \sqrt{1^2+1^2}$

$$= \sqrt{2}$$

$$\therefore 1-i = \sqrt{2} \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i \right)$$

$$= \sqrt{2} \left(\cos\left(-\frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4}\right) \right)$$

(ii) $(1-i)^5 = \sqrt{2}^5 \left[\cos\left(-\frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4}\right) \right]^5$

$$= 4\sqrt{2} \left[\cos\left(-\frac{5\pi}{4}\right) + i \sin\left(-\frac{5\pi}{4}\right) \right]$$

$$= 4\sqrt{2} \left[\cos\left(-\pi - \frac{\pi}{4}\right) + i \sin\left(-\pi - \frac{\pi}{4}\right) \right]$$

$$= 4\sqrt{2} \left[\cos\left(\pi - \frac{\pi}{4}\right) + i \sin\left(\pi - \frac{\pi}{4}\right) \right]$$

$$= 4\sqrt{2} \left[\cos\frac{3\pi}{4} + i \sin\frac{3\pi}{4} \right]$$

(iii)

$$(1-i)^5 = 4\sqrt{2} \left[\cos\left(\pi - \frac{\pi}{4}\right) + i \sin\left(\pi - \frac{\pi}{4}\right) \right]$$

$$= 4\sqrt{2} \left[-\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right]$$

$$= -4 + 4i$$

~~✓~~ A
C

$$\frac{1}{2} + \frac{1}{2} = 1$$

9

$$a_2 = 100$$

$$h^2 = 75$$

810

$$b = 5\sqrt{3}$$

75

$$\therefore b^2 = a^2(1 - e^2)$$

$$75 = 100(1 - e^{-2})$$

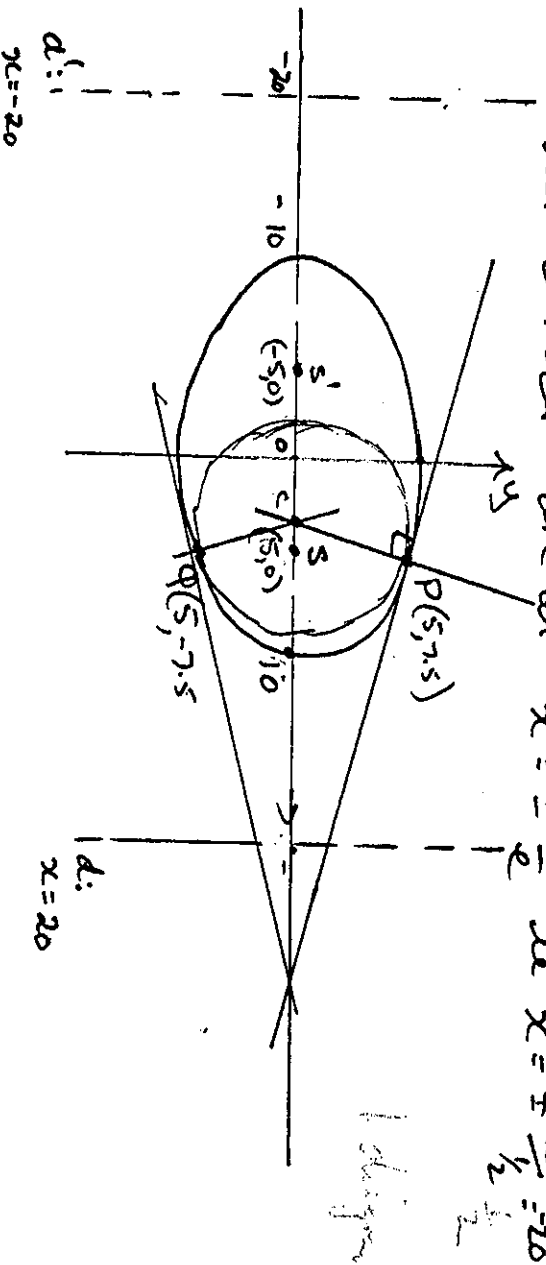
$$\frac{75}{100} = 1 - e^2$$

$$\frac{25}{100} = \frac{1}{4}$$

२
 "
 ०/५
 "
 २५

0 < \epsilon < 1.

$\therefore$ Foci are at $(\pm ae, 0)$ i.e. $(5, 0)$ & $(-5, 0)$
directrices are at $x = \pm \frac{a}{e}$ i.e. $x = \pm \frac{10}{\frac{1}{2}} = \pm 20$



③

$$\frac{2x}{100} + \frac{2y}{75} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-2x}{100} \times \frac{75}{2y}$$

$$\frac{-3x}{5}$$

$$\text{when } x = 5, y = 7.5, \frac{dy}{dx} = \frac{-3 \times 5}{4 \times 7.5} = \frac{-15}{30}$$

gradient of normal is ∇n

2

$$\therefore \text{Egn of Normal at } P \text{ is } y - 7.5 = 2(x - 5) \text{ i.e.}$$

$$\therefore y - 7.5 = 2x - 10$$

$$2x - y - 2.5 = 0$$

$$x^2 - 4x - 29 = 0$$

(iii)

Because of the symmetry of the ellipse & the points P & Q , the normal intersect on x -axis when $y=0$, $4x-5=0 \Rightarrow x = \frac{5}{4}$

$$x = \frac{5}{4} \quad \text{on } C \left(\frac{5}{4}, 0 \right)$$

distance PC is

$$\begin{aligned} & \sqrt{\left(5 - \frac{5}{4}\right)^2 + (7.5 - 0)^2} \\ &= \sqrt{\left(\frac{15}{4}\right)^2 + 7.5^2} \\ &= \sqrt{\frac{225}{16} + \frac{225}{4}} = \sqrt{70\frac{5}{8}} = \sqrt{\frac{1125}{16}} \end{aligned}$$

$\therefore$ Eqn circle is

$$\left(x - \frac{5}{4}\right)^2 + y^2 = \frac{1125}{16}$$

Q3.

$$A(x) = x^4 + ax^2 + bx + 36$$

double root

$$A(2) = 16 + 4a + 2b + 36 = 0$$

1/2

$$A'(x) = 4x^3 + 2ax + b$$

angle root

$$A'(2) = 32 + 4a + b = 0$$

$$\therefore 4a + 2b + 5a = 0 \quad \text{--- (1)}$$

$$4a + b + 32 = 0 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2} \quad b + 20 = 0$$

$$b = -20$$

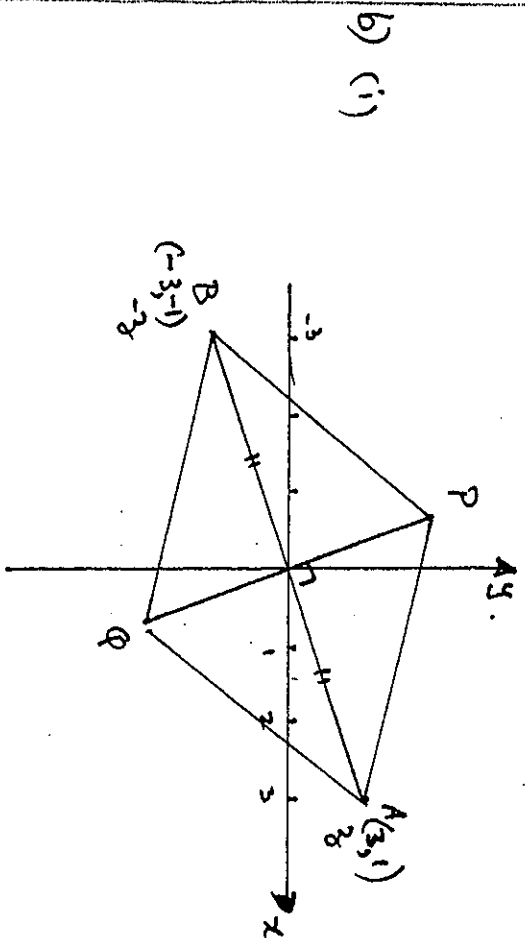
$$\text{Sub into } \textcircled{1}: 4a + -40 + 5a = 0$$

$$4a + 12 = 0$$

$$4a = -12$$

$$a = -3$$

$$\therefore a = -3, \quad b = -20$$



(ii) Since $APBQ$ is a rhombus then
 if $P(w)$ and $Q(-w)$
 and $PQ \perp AB$
 and $PO = OQ$, $AO = OB$ $\frac{1}{2}$

now $A \rightarrow P$ by the rotation anticlockwise
 and an enlargement by k $\frac{1}{2}$
 thus $kx(3+i) = 3ki + ki^2$

Then P is $(-k, 3k)$ and Q is $(k, -3k)$ $\frac{1}{2}$
 Now, $A = \frac{1}{2} \cdot AB \cdot PQ$
 $150 = \frac{1}{2} \times \sqrt{40} \times \sqrt{40} k$
 $300 = 40k$
 $\frac{300}{40} = k$
 $7.5 = k$ $\frac{1}{2}$

$$PQ = \sqrt{(k - (-k))^2 + (-3k - 3k)^2}$$

$$= \sqrt{4k^2 + 36k^2}$$

$$= \sqrt{40} k$$

$\therefore P$ is $(-7.5, 22.5)$
 Q is $(7.5, -22.5)$ $\frac{1}{2}$

c) (i)

$$\frac{PS}{PM} = e$$

$$PS = e PM$$

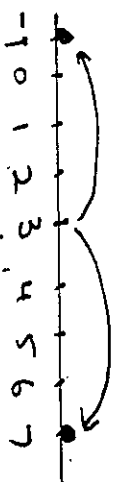
and $\frac{PS'}{PM'} = e$

$$PS' = e PM'$$

$$PS + PS' = e(PM + PM')$$

$$= e \left[\left(\frac{a}{e} + \frac{a}{e} \right) \right] = 2a$$

(ii)

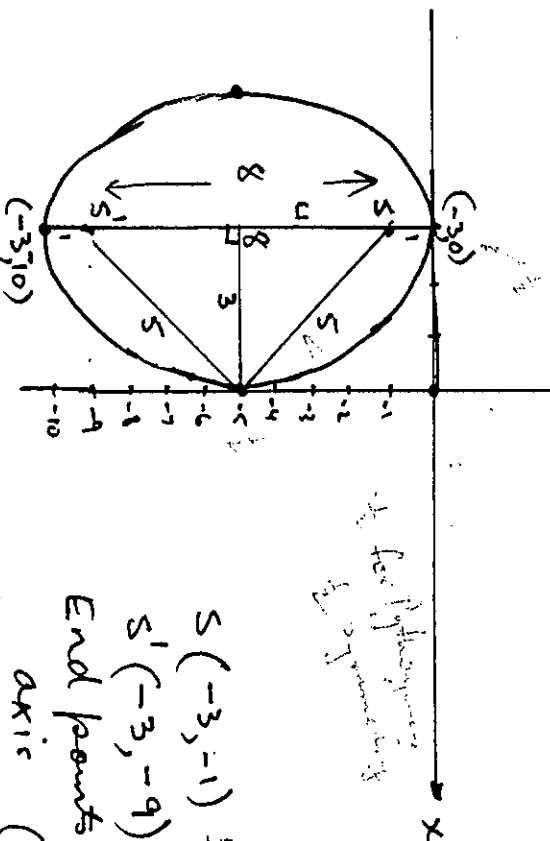


"The distance of x from 3 is 4 units"

$$|3 + 3 + i| + |3 + 3 + 9i| = 10$$

$$|3 - (-3 - 9)| + |3 - (-3 - 9i)| = 10$$

The distance of 3 from $(-3, -1)$ added to distance from $(-3, -9)$ is 10 units



$$PS + PS' = 10 \frac{1}{2}$$

$$S(-3, -1) \frac{1}{2}$$

$$S'(-3, -9) \frac{1}{2}$$

End points of major axis $(-3, 0)$ and $(-3, 10)$

End point minor axis

$$(0, -5) \text{ and } (-6, -5)$$

Centre of ellipse $(-3, -5) \frac{1}{2}$

(iv) $-\pi \leq \arg z \leq -\frac{\pi}{2}$ is range of values

OR alternatively - although not using the method told "

(iii)

$$|3+3+i| + |3+3+9i| = 10 \Rightarrow |(x+3)+i(1+y)| + |(x+3)+i(y+9)| = 10$$

$$\sqrt{(x+3)^2 + (1+y)^2} + \sqrt{(x+3)^2 + (y+9)^2} = 10$$

$$\sqrt{(x+3)^2 + (1+y)^2} = 10 - \sqrt{(x+3)^2 + (y+9)^2}$$

$$(x+3)^2 + (1+y)^2 = 100 + (x+3)^2 + (y+9)^2 - 20\sqrt{(x+3)^2 + (y+9)^2}$$

$$20\sqrt{(x+3)^2 + (y+9)^2} = 100 + (y+9)^2 - (1+y)^2$$

$$20\sqrt{(x+3)^2 + (y+9)^2} = 100 + [(y+9) + (1+y)][(y+9) - (1+y)]$$

$$20\sqrt{(x+3)^2 + (y+9)^2} = 100 + (2y+10)(8)$$

$$5\sqrt{(x+3)^2 + (y+9)^2} = 25 + 2(2y+10)$$

$$5\sqrt{(x+3)^2 + (y+9)^2} = 25 + 4y + 20$$

$$5\sqrt{(x+3)^2 + (y+9)^2} = 4y + 45$$

$$25[(x+3)^2 + (y+9)^2] = 16y^2 + 360y + 2025$$

$$25(x+3)^2 + 25(y+9)^2 = 16y^2 + 360y + 2025$$

$$25(x+3)^2 + 9y^2 + 90y = 0$$

$$25(x+3)^2 + 9(y^2 + 10y + 25) = 225$$

$$25(x+3)^2 + 9(y+5)^2 = 225$$

$$\frac{25(x+3)^2}{225} + \frac{9(y+5)^2}{225} = 1$$

$$\frac{(x+3)^2}{(15/5)^2} + \frac{(y+5)^2}{(5/3)^2} = 1$$

$$\frac{(x+3)^2}{9} + \frac{(y+5)^2}{25} = 1$$

$$a=3, b=5$$

$$a^2 = b^2(1-e^2)$$

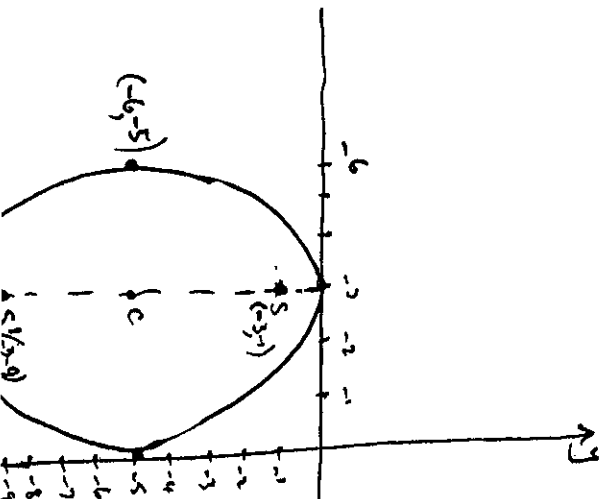
$$9 = 25(1-e^2)$$

$$\frac{9}{25} = 1-e^2$$

$$e^2 = 1 - \frac{9}{25} = \frac{16}{25}$$

$$e = \frac{4}{5}$$

$$be = 5 \times \frac{4}{5} = 4$$



(ii) ~~also~~ since M, midpt of BC is centre of circle through C, X and B.
then $MC = MB = MX$ (= radii same circle)
then $\triangle MBX$ is isosceles.
Thus, $\angle MBX = \angle MXB$ (= base $\angle$'s opp = sides, isosceles $\triangle$)

Then $\angle BXM = x^\circ$ (proved above)
 $\angle BCX = 90^\circ - x^\circ$ ($\triangle BCX$ right angled at X ; $\angle XCB$ complementary angle to $\angle CBX$)

Thus, in ΔDXN ,
 $\angle DXN = x^\circ$. $\angle M \times B$ (vertically)
 $\angle XND + \angle DXN + \angle XDN = 180^\circ$
 $\angle XND + x^\circ + 90^\circ - x^\circ = 180^\circ$
 $\angle XND = 90^\circ$

$\Delta \times \nabla$

b) (i)

$$\cos \theta = \frac{OT}{OM} = \frac{a}{OM}$$

$$\therefore OM = \frac{a}{\cos \theta}$$

$$\therefore OM = a \sec \theta$$

$$\text{Now } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\therefore \frac{a^2 \sec^2 \theta}{a^2} - \frac{y^2}{b^2} = 1$$

$$\sec^2 \theta - 1 = \frac{y^2}{b^2}$$

$$\tan^2 \theta = \frac{y^2}{b^2}$$

$$b^2 \tan^2 \theta = y^2$$

$$b \tan \theta = y$$

$$\therefore P(a \sec \theta, b \tan \theta)$$

(ii) $P(a \sec \theta, b \tan \theta)$ and $Q(a \sec \phi, b \tan \phi)$

Eqn of chord PQ is

$$\frac{y - b \tan \theta}{x - a \sec \theta} = \frac{b \tan \phi - b \tan \theta}{a \sec \phi - a \sec \theta} \times \frac{\sec \theta \cos \phi}{\cos \theta \cos \phi}$$

$$\frac{y \cos \theta - b \sec \theta}{x \cos \theta - a} = \frac{b (\sec \phi \cos \theta - \sec \theta \cos \phi)}{a (\cos \theta - \cos \phi)}$$

Now $\phi = \frac{\pi}{2} - \theta$,

$$\therefore \frac{y \cos \theta - b \sec \theta}{x \cos \theta - a} = \frac{b [\sec(\frac{\pi}{2} - \theta) \cos \theta - \sec \theta \cos(\frac{\pi}{2} - \theta)]}{a [\cos \theta - \cos(\frac{\pi}{2} - \theta)]}$$

$$= \frac{b [\cos \theta \cos \theta - \sec \theta \sin \theta]}{a [\cos \theta - \sin \theta]}$$

$$= \frac{b}{a} \frac{(\cos \theta - \sin \theta)(\cos \theta + \sin \theta)}{\cos \theta - \sin \theta}$$

$$\frac{y \cos \theta - b \sec \theta}{x \cos \theta - a} = \frac{b(\cos \theta + \sin \theta)}{a}$$

$$ay \cos \theta - ab \sec \theta = bx(\cos \theta + \sin \theta) \cos \theta - ab \cos \theta \sin \theta$$

$$ay \cos \theta - ab \sec \theta = b x (\cos \theta + \sin \theta) \cos \theta - ab \cos \theta \sin \theta$$

$$ay \cos \theta = bx(\cos \theta + \sin \theta) \cos \theta - ab \cos \theta$$

$$\therefore ay = bx(\cos \theta + \sin \theta) - ab$$

(iii) $y = \frac{b}{a}(\cos \theta + \sin \theta)x - b$
 $\therefore$ all such straight lines pass through $(0, -b)$

(iv) as $\theta \rightarrow \frac{\pi}{2}$, $y = \frac{b}{a}(1 + 0^+)x - b$

ie $y = \frac{b}{a}x - b$
 as asymptotes are $y = \pm \frac{b}{a}x \therefore$ this line is parallel to the asymptote $y = \frac{b}{a}x$ (same gradient)

Q5.

a)

$$P(z) = z^4 + bz^2 + a$$

(i)

$$P'(z) = 4z^3 + 2bz$$

$$P'(z) = 4z^3 + 2bz \\ = 2z(2z^2 + b)$$

$$P'(-z) = 4(-z)^3 + 2b(-z) \\ = -4z^3 - 2bz \\ = -2z(2z^2 + b)$$

$$\therefore P'(z) = -P'(-z) \\ \therefore \text{fn } P(z) \text{ is odd.}$$

(ii)

$$(z - \alpha) \text{ is a factor of } P'(z) \\ (z - \alpha)^2 \text{ " " " } P(z)$$

Thus since $P'(z)$ is odd then $-\alpha$ is a root of $P'(z)$ to match root $x = \alpha$.
ie $(z + \alpha)$ is a factor of $P'(z)$
+ so, $(z + \alpha)^2$ is a factor of $P'(z)$.

(iii)

$$P(z) = (z - \alpha)^2 (z + \alpha)^2 \\ = [(z - \alpha)(z + \alpha)]^2 \\ = [z^2 - \alpha^2]^2$$

$$= z^4 - 2\alpha^2 z^2 + \alpha^4$$

Thus equating coefficients.

$$-2\alpha^2 = b$$

and

$$\alpha^4 = a$$

$$\alpha^2 = \frac{b}{-2}$$

$$(\alpha^2)^2 = a$$

$$\left(\frac{b}{-2}\right)^2 = a$$

(iv)

$$\text{if } \alpha = \sqrt{3}i, \alpha^2 = 3i^2 \\ = -3$$

$$\frac{b^2}{4} = a$$

$$\therefore \alpha^2 = \frac{b}{-2} = -3$$

$$\therefore b = 6$$

(v) For $P(z)$ to have real roots $\alpha^2 = \frac{b}{-2} > 0 \therefore b < 0$

$$b) (i) \sin(A+B) - \sin(A-B) = \sin A \cos B + \sin B \cos A - [\sin A \cos B - \sin B \cos A]$$

$$= \sin A \cos B + \sin B \cos A - \sin A \cos B + \sin B \cos A$$

$$= 2 \sin B \cos A$$

$$= 2 \cos A \sin B$$

To Prove that

$$(ii) \cos x + \cos 2x + \cos 3x + \dots + \cos nx = \frac{\sin(n+\frac{1}{2})x - \sin\frac{1}{2}x}{2 \sin\frac{1}{2}x}$$

$$\text{Test } n=1$$

$$\text{LHS} = \cos x$$

$$\text{RHS} = \frac{\sin(x+\frac{1}{2}x) - \sin\frac{1}{2}x}{2 \sin\frac{1}{2}x}$$

$$= \frac{\sin(x+\frac{1}{2}x) - \sin(x-\frac{1}{2}x)}{2 \sin\frac{1}{2}x} \text{ From (i)}$$

$$= \frac{2 \cos x \sin\frac{1}{2}x}{2 \sin\frac{1}{2}x}$$

$$= \cos x$$

$$\therefore \text{LHS} = \text{RHS}$$

True for $n=1$.

Assume true for $n=k$,

$$\sin \cos x + \cos 2x + \dots + \cos kx = \frac{\sin(k+\frac{1}{2})x - \sin\frac{1}{2}x}{2 \sin\frac{1}{2}x}$$

Prove true for $n=k+1$,

$$\cos x + \cos 2x + \dots + \cos kx + \cos(k+1)x = \frac{\sin(k+\frac{1}{2})x - \sin\frac{1}{2}x}{2 \sin\frac{1}{2}x} + \frac{\sin(k+\frac{1}{2})x + \sin(k+1-\frac{1}{2})x}{2 \sin\frac{1}{2}x}$$

$$= \frac{\sin(kx+\frac{1}{2}x) - \sin\frac{1}{2}x + [\sin(k+\frac{1}{2})x - \sin(k+1-\frac{1}{2})x]}{2 \sin\frac{1}{2}x}$$

$$= \frac{\sin(k+\frac{1}{2})x - \sin\frac{1}{2}x + \sin(k+\frac{1}{2})x - \sin(k+\frac{1}{2})x}{2 \sin\frac{1}{2}x}$$

$$= \frac{\sin(k+\frac{1}{2})x - \sin\frac{1}{2}x}{2 \sin\frac{1}{2}x}$$

is true for all n +ve integers.